

How to Balance Chemical Equations

Expert

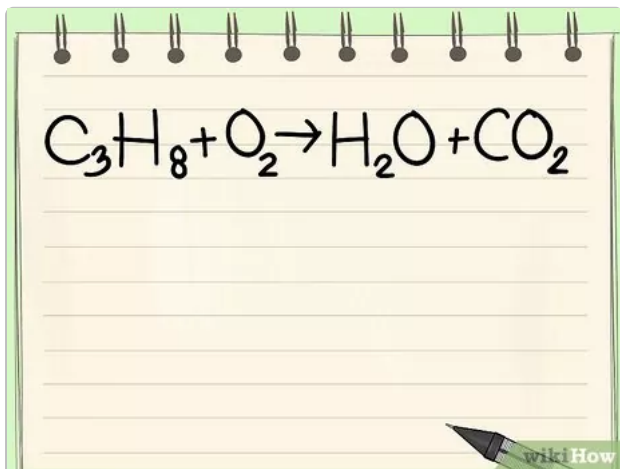
Co-Authored

A chemical equation is a written symbolic representation of a chemical reaction. The reactant chemical(s) are given on the left-hand side and the product chemical(s) on the right-hand side. The two are connected with an arrow leading from the left to the right, symbolizing the reaction. The law of conservation of mass states that no atoms can be created or destroyed in a chemical reaction, so the number of atoms that are present in the reactants has to balance the number of atoms that are present in the products. Follow this guide to learn how to balance

chemical equations differently.

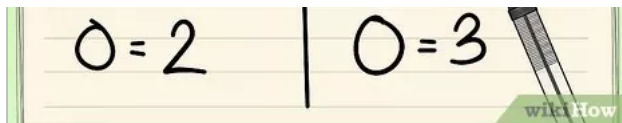
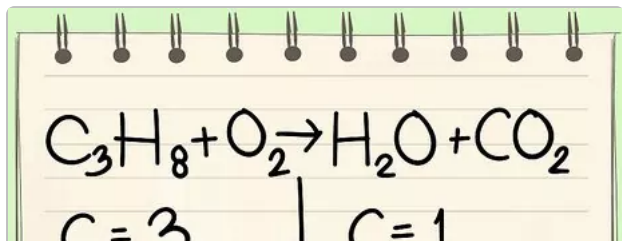
Method One of Two:

Traditional Balance



- 1 Write down your given equation.** For this example, you will use:

- $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$
- This reaction occurs when propane (C_3H_8) is burned in the presence of oxygen to produce water and carbon dioxide.

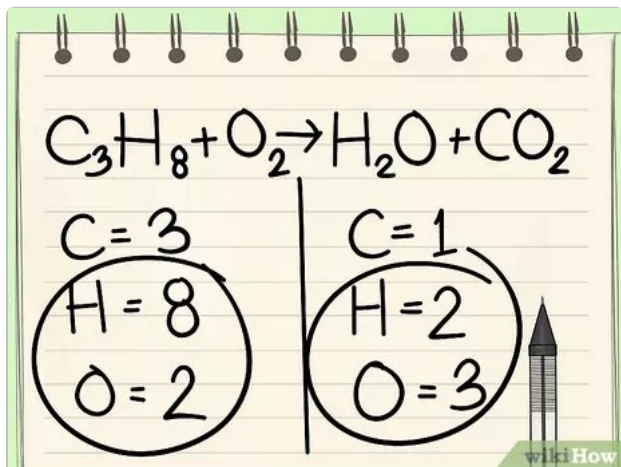


Write down the number of atoms

2 per element. Do this for each side of the equation. Look at the subscripts next to each atom to find the number of atoms in the equation. When writing it out, it's a good idea to connect it back to the original equation, noting how each element appears.

- For example, you have 3 oxygen atoms on the right side, but that total results from addition.
- Left side: 3 carbon (C₃), 8 hydrogen (H₈) and 2 oxygen (O₂).
- Right side: 1 carbon (C), 2 hydrogen (H₂) and 3 oxygen

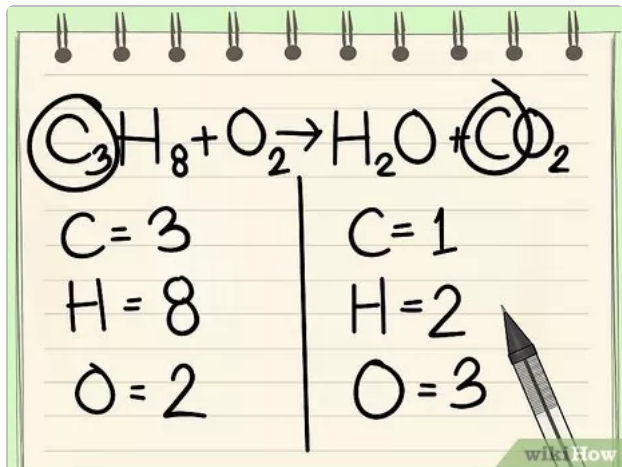
(0 + 02).



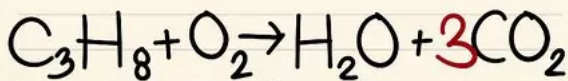
3 Save hydrogen and oxygen for last, as they are often on both sides. Hydrogen and oxygen are both common in molecules, so it's likely that you'll have them on both sides of your equation. It's

best to balance them last.

- You'll need to recount your atoms before balancing the hydrogen and oxygen, as you'll likely need to use coefficients to balance the other atoms in the equation.



4 Start with single elements. If you have more than one element left to balance, select the element that appears in only a single molecule of reactants and in only a single molecule of products. This means that you will need to balance the carbon atoms first.



$$\text{C} = 3$$

$$\text{H} = 8$$

$$\text{O} = 2$$

$$\text{C} = \cancel{1} 3$$

$$\text{H} = 2$$

$$\text{O} = \cancel{2} 7$$

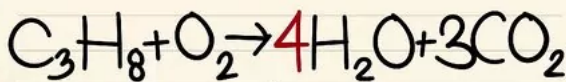
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5 Use a coefficient to balance the single carbon atom. Add a coefficient to the single carbon atom on the right of the equation to balance it with the 3 carbon atoms on the left of the equation.

- $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{H}_2\text{O} + 3\text{CO}_2$
- The coefficient 3 in front of

carbon on the right side indicates 3 carbon atoms just as the subscript 3 on the left side indicates 3 carbon atoms.

- In a chemical equation, you can change coefficients, but you must never alter the subscripts.



$$\text{C} = 3$$

$$\text{H} = 8$$

$$\text{O} = 2$$

$$\text{C} = 3$$

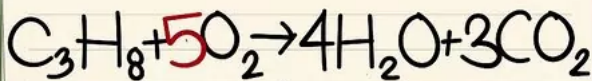
$$\text{H} = \cancel{2} 8$$

$$\text{O} = \cancel{7} 10$$

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6 **Balance the hydrogen atoms next.** Since you have balanced all atoms besides the hydrogen and oxygen, you can address the hydrogen atoms. You have 8 on the left side. So you'll need 8 on the right side. Use a coefficient to achieve this.

- $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow 4\text{H}_2\text{O} + 3\text{CO}_2$
- On the right side, you now added a 4 as the coefficient because the subscript showed that you already had 2 hydrogen atoms.
- When you multiply the coefficient 4 times by the subscript 2, you end up with 8.
- The other 6 atoms of oxygen come from 3CO_2 . ($3 \times 2 = 6$ atoms of oxygen + the other 4 = 10)



$$\text{C} = 3$$

$$\text{H} = 8$$

$$\text{O} = \cancel{2} 10$$

$$\text{C} = 3$$

$$\text{H} = 8$$

$$\text{O} = 10$$

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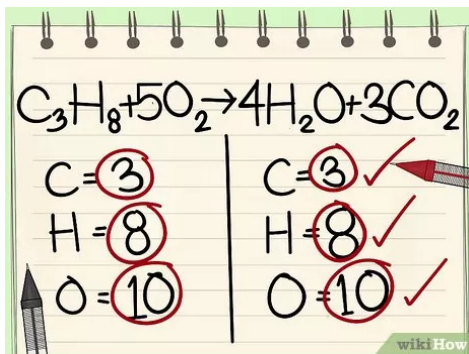
7 Balance the oxygen atoms.

Remember to account for the coefficients that you've used to balance out the other atoms.

Because you've added coefficients to the molecules on the right side of the equation, the number of oxygen atoms has changed. You now have 4 oxygen atoms in the

water molecules and 6 oxygen atoms in the carbon dioxide molecules. That makes a total of 10 oxygen atoms.

- Add a coefficient of 5 to the oxygen molecule on the left side of the equation. You now have 10 oxygen atoms on each side.
- $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 4\text{H}_2\text{O} + 3\text{CO}_2$.



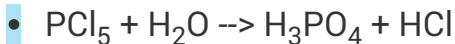
- The carbon, hydrogen, and oxygen atoms are balanced. Your equation is complete.

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Method Two of Two:

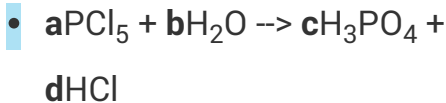
Algebraic Balance

1 Write down the given equation. For this example, we will use:



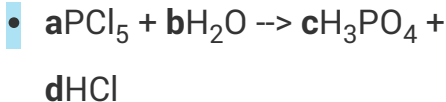
Assign a letter to each substance.

2



3

Check the number of each element found on both sides, and set those equal to each other.



- On the left side, there are $2b$ atoms of hydrogen (2 for every molecule of H_2O), while on the right side, there are $3c+d$ atoms of hydrogen (3 for every molecule of H_3PO_4 and

1 for every molecule of HCl). Since the number of atoms of hydrogen has to be equal on both sides, $2b$ must be equal to $3c+d$.

- Do this for every element.
 - P: $a=c$
 - Cl: $5a=d$
 - H: $2b=3c+d$

4 Solve this system of equations to get the numeric value for the coefficients. Since there are more variables than equations, there are multiple solutions. You must find the one

where every variable is in its smallest, non-fractional form.

- To quickly do this, take one variable and assign a value to it. Let's make $a=1$. Then start solving the system of equations to get the following values:
- Since P: $a = c$, we know that $c = 1$.
- Since Cl: $5a = d$, we know that $d = 5$
- Since H: $2b = 3c + d$, we can calculate b like this:
 - $2b = 3(1) + 5$
 - $2b = 3 + 5$
 - $2b = 8$

- $b=4$
- This shows us the values are as follows:
 - $a = 1$
 - $b = 4$
 - $c = 1$
 - $d = 5$

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How did the oxygen became 7

when 3×2 is 6? Do you always add one?

wikiHow Contributor

Community Answer

- You get $3 \times 2 = 6$ oxygen atoms in the carbon dioxide, but at this stage there is also 1 oxygen atom in the water. So, $6 + 1 = 7$.

• Question

How do I understand this information?

wikiHow Contributor

Community Answer

- Read the steps carefully and keep a notebook and a

pen with you during reading the steps. After reading the steps, write each step one by one "in your own words", noting it in your notebook one by one and using the picture to enhance your comprehension.

- Question

How can I balance $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$?

wikiHow Contributor

Community Answer

- This equation is already balanced when using the first method (Traditional

balance) in the article.

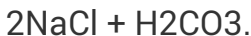
- Question

How can I balance $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{CO}_3$?

wikiHow Contributor

Community Answer

- First multiply NaCl by 2, since you have 2 Na on the reactant side. $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{CO}_3$. Then, since there are now 2 Cl in the product side, multiply HCl by 2. This also has the effect of balancing the hydrogen, and the equation is now balanced as $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow$



- Question

What causes the color of copper sulphate to change when an iron nail is dipped into it?

wikiHow Contributor

Community Answer

- When iron (Fe) and copper sulphate (CuSO_4) solution react, they undergo a single displacement reaction, also known as a substitution reaction, to form solid copper (Cu) and aqueous iron sulphate (FeSO_4). The iron can be solid or aqueous but the

copper sulphate must be aqueous in order to facilitate the reaction.

- Question

How do I balance $\text{HgO} = \text{Hg} + \text{O}_2$?

wikiHow Contributor

Community Answer

- $\text{HgO} \rightarrow \text{Hg} + \text{O}_2$. Multiply the left side by 2 to get 2 Oxygen. $2\text{HgO} \rightarrow \text{Hg} + \text{O}_2$. Then multiply Hg on the right by 2. Et voilà! We get the balanced $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$.

- Question

On step 7 it says you now have 6 oxygen atoms in the carbon

dioxide molecule do I times the 3 to get 6?

wikiHow Contributor

Community Answer

- Yes. You multiply the subscript by the number that was put out in front. So CO_2 has one carbon and two oxygen atoms (pretend the 2 is subscript), while $3(\text{CO}_2)$ has 3 carbon and six oxygen atoms. If it helps (and again, pretend that all the numbers in this equation are subscript), $\text{CO}_2 = \text{C}_1\text{O}_2$. You always multiply the number in

front by the subscript and if there is none, it means that the subscript is equal to 1.

- Question

How can we assume $a=1$. It can be anything right?

wikiHow Contributor

Community Answer

- It is a balanced equation. We don't assume anything, the writer chose to give you the answer of "a" for use in this example. If you did not know $a=1$, then it could theoretically be anything.

- Question

Why does $a = c$ for the algebraic method?

wikiHow Contributor

Community Answer

- According to the law of conservation of mass, the amount of P will be the same on both sides. So the amount of P as a reactant (a), will be the same as the amount of the product of P (c).

- Question

What is the action of diluted nitric acid on magnesium?

wikiHow Contributor

Community Answer

- Magnesium will react with diluted nitric acid to produce magnesium nitrate and hydrogen gas.

Tips

- If you're stuck, you can type the equation into the online balancer to balance it. Just remember that you won't have access to an online balancer when you're taking an exam, so don't become dependent on it.
- Remember to simplify! If all of your coefficients can be divided

by the same number, do so to get the simplest result.

Warnings

- To get rid of fractions, multiply the entire equation (both the left and right sides) by the number in the denominator of your fraction.
- Never use fractions as coefficients in a chemical equation--you never make half of a molecule or half of an atom in a chemical reaction.
- During the balancing process, you may use fractions to assist

you, but the equation is not balanced as long as there are still coefficients using fractions.

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About This Article

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Bess Ruff

MA in Environmental Science and
Management

This version of How to Balance Chemical Equations was expert co-authored by Bess Ruff on July 20, 2018. Learn more...

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Article Summary

- 1.** Write down the number of atoms in each element in the equation.
- 2.** Start with the element that appears in a single molecule of reactants and in a single molecule of products.
- 3.** Balance the element by adding a coefficient to it on the right so it has the same number of atoms as it does on the left.

4. Repeat with the remaining elements in the equation. When you're finished, each element on the left side of the equation should have the same atoms as the corresponding element on the right.

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- A

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more

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